Problem-solving and Decision-making Techniques

Problem-solving and decision-making are key activities within project management. In fact many projects will be set in motion to solve a problem and you will be making decisions through to completion.

There are numerous approaches to problem-solving and decision-making. These can be highly structured or deliberately minimal. The key is to select the most appropriate one for your purposes. In this information sheet, we outline three popular problem-solving techniques and then go on to look at a number of decision-making tools.

Problem-solving

Problem-solving Steps:

If you are faced with a problem, it's sometimes difficult to know where to begin. Using the following five-step approach may help to get you started.[1]

- First, **define the problem**. Being clear about the nature of the problem will set the course of how you intend to solve it.
- Next, **determine the causes**. If you look at the problem as a gap between where you are now and where you want to be, it can help you determine reasons for the gap along with any obstacles that might be in the way.
- Then, **generate ideas**. Now that the first two stages are complete, you can start to use your imagination to generate some possible solutions.
- After that, **select the best solution**. This is about making a decision and involves an evaluation of each possible solution. We offer some ways that you could do this later in this information sheet.
- Finally, **take action**. This may result in a series of actions that may best be set out in an action plan.

Problem-solving Techniques

There are many different techniques that you can use for problem solving. Here, we outline three popular approaches: brainstorming, mind-mapping and the fishbone diagram.

- **Brainstorming**

This method of problem solving can be undertaken individually or in a group setting. The process requires as many ideas as possible to be generated, no matter how outlandish they might appear. Once all the suggestions are on the table, and not before, then each will be evaluated against the others. It is suggested that this type of 'free thinking environment' has the benefit of unlocking creativity.

[1] https://www.project-management-skills.com/problem-solving-techniques.html

In a group setting, as everyone is invited to participate, it can improve team cohesion and lead to increased self-esteem. However, a group brainstorm must be carefully facilitated and it can be a lengthy process.

- **Mind-mapping**

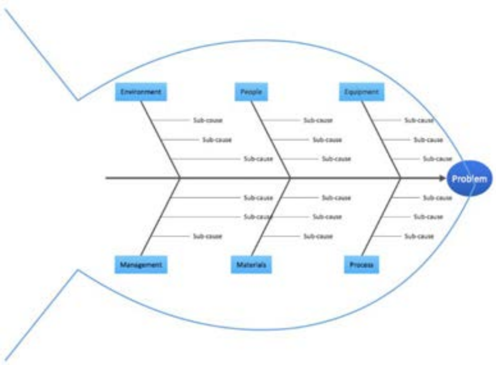
A mind-map is a visual method of organising information or mapping out ideas. It's a process that enables you to see the bigger picture through the prioritisation and association relating to elements of a problem. This process can help a project manager to organise their thoughts and can make more complex problems easier to understand.

There are no real rules to creating a mind map and it can be created in different shapes and colours as a way of stimulating the brain. However, for some project managers, working with a visual medium might not come naturally, so they may do better with a more structured approach, such as the one we will cover next.

- **The Fishbone Diagram**

The fishbone or Ishikawa diagram, named after its originator, is a visual *cause and effect* approach that looks to the possible reasons why a problem occurred in order to discover its root cause. The approach pulls together some of the features of brainstorming and mind mapping. As the name suggests, the diagram looks like a fish skeleton with the problem at its head and the causes as the bones of its spine.

To complete the diagram, you start with the head. Here you write down the exact nature of the problem. This might include what the problem is, who is involved and where it is. Traditionally, the causes are on the left and the effect on the right, although you could choose to present the diagram the other way round.
Then you identify the factors that could be part of the problem. For example, these could include systems, process, equipment, site, external forces or people. You enter your causes on the bones that connect to the spine of the fish. Then taking each factor in turn, you think of the possible causes relating to that factor. These will form further lines, coming off the 'bones'.
The outline of your diagram may look something like this:



This is a useful visual tool which is easy to understand and should help you to find the root cause of the problem.

Decision-making

Whatever the project, it will always involve some form of decision-making. In this section, we look at two main approaches along with some popular decision-making models.

Decision-making approaches may be loosely grouped into two main categories: non-systematic and systematic.

Non-systematic Decision-making Approaches

Non-systematic approaches rely on shortcuts of one form or another, and are often intuitive, rather than consciously set out.

If you look at the decisions you make over the course of a working day, you'll find that many of them are non systematic.

The most common such approaches are:

1. **Making a random choice**: this method tends to work well with relatively unimportant decisions. Where the stakes are higher, the wrong choice could have serious consequences. If, for example, you're deciding where to invest money, making a random choice wouldn't necessarily be your best move.

2. **Imitating others**: for example, this could be following tradition. This approach benefits from the experiences of others, and can help hasten decisions. However, it means missed opportunities to try untested, but possibly better options. Additionally, what has worked once before may not work again in a different context.

3. **Going by experience**: this can lead to better choices on important issues that we are familiar with. However, we may be overconfident in our expertise, remember things inaccurately, or fail to detect our own biases. All of these may lead to the wrong decision.

4. **Listening to intuition**: intuition is defined as our ability to understand something using our instincts, rather than conscious reasoning.[2] It can be a powerful shortcut to decision-making but could be subject to personal bias. Academic Eugene Sadler-Smith suggests that 'people need to have a certain level of experience to make effective use of intuition[3]. He argues that experience can 'test' our sense of intuition, to ensure it isn't biased or otherwise illogical.

[2] Oxford Compact English Dictionary.
[3] Cited by Morrish, J. (2012) in 'Should You Trust Your Gut Feelings?' *Management Today*, 1 November.

Systematic Decision-making Approaches

When you approach a decision systematically, you will be following a logical and orderly process that has been developed in advance. This doesn't automatically mean that it needs to be complex, but by taking an organised approach, you will be less likely to miss critical factors which should lead to better decisions. We outline five systematic approaches here.

- **PMI**

The Plus, Minus and Interesting, or PMI model. This model was created by business psychologist Edward de Bono. It divides actions by whether their outcomes are positive, negative, or neutral (but in some way interesting and relevant) to the choice.

PMI is easy to use by individuals or groups, and can be applied quickly. However, the simplicity of the model can also be a weakness. The structure is so loose that it can reinforce existing preferences instead of testing them.

For example, there is no requirement to consider your objectives when using PMI - but these are important to ensure the outputs from the model are fit for purpose; if the model is used in a group setting and more forceful individuals are present, the voices of less confident contributors may not be heard.

- **Cost Benefit Analysis (CBA)**

To use CBA, a manager assigns financial costs and benefits to the competing actions - this kind of quantification can simplify decision-making.

However, some costs and benefits can be highly subjective. For example, how do we assign value to the beauty of the landscape when a project involves choosing between two locations for a housing development?

- **Grid Analysis**

This method sets out the possible options and tests them against a range of criteria. So, for example, a firm wishing to move its head office might assess potential locations against their cost, the transport links, and location of business partners.

However, decision-making isn't always so clear cut. One choice can lead to more choices, and therefore more decisions - which might result in several outcomes.

This situation calls for a more sophisticated decision-making tool - the decision tree.

- **Decision Trees**

Decision trees are diagrams that represent each choice as a branch of a tree. Any given decision may result in a range of possible outcomes; this may prompt further decisions, resulting in more branches. As these can get quite complicated to compare, each outcome can be given a cost. It may also be assigned a probability, expressed as a decimal or a percentage**.** The financial costs or benefits of each path through the decision tree can then be calculated, based on the likelihood of each outcome.[4]

For this reason, you may sometimes hear the approach described as 'probability analysis,' and decision trees as 'probability trees.' A worked example of how a decision tree might work in practice is available in the accompanying document: 'Decision Trees Information Sheet'.

- **Kepner and Tregoe's Process**

Kepner and Tregoe's Rational Decision-making model was first published in 1965. It's a process that can not only help you make rational choices, but also develop the options for them. It aimed to encourage more objective decision-making at a time when many business decisions were based on the subjective desires of the powerful.[5] Kepner and Tregoe's model provides a simple and rational framework for making decisions when a clear goal can be identified. It supports transparency and impartiality.

This decision-making process is made up of a number of steps. It might look rather complex at first glance, but by making rational choices, the best decision will be reached.

1. Prepare a decision statement which includes the desired result.
2. Define the objectives in terms of strategic requirements and operational objectives.
3. Classify the objectives into 'must have', or those that are essential, and 'want to have', or things that would be nice but aren't critical.
4. List alternatives to generate as many possible courses of action as possible.
5. Eliminate any that are not classified as 'must have', then score each alternative by assigning relative weights on a scale of 1 to 10. The most important want receives a 10, with the remaining objectives weighted in relation to this.
6. Choose the top two or three alternatives and consider potential problems or negative effects of each.
7. Consider each alternative against all of the negative effects.
8. Analyse the weighted score against the adversity rating for each and choose the high-scoring one.
9. You should now have found your winning option.

Next time you have to solve a project-related problem or need to make a decision, think about using some of the techniques we have shown you and see if they help.

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[4] Law, J. (ed) *Oxford Dictionary of Business and Management* (Oxford: Oxford University Press)
[5] Kepner, C.H. & Tregoe, B.B (2013) *The New Rational Manager* (Princeton: Princeton Research Press)